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Modeling the Impulse Response of an Underwater Wireless Optical Communication System

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***Abstract*—The Channel Impulse Response (CIR) for an Underwater Wireless Optical Communication (UWOC) channel is estimated by employing the Monte-Carlo numerical simulation technique. Realistic marine optical and system parameters are considered based on the reported literature. The estimated CIR is further modeled empirically to include the effects of optical and system parameters on the channel impulse response. The mathematical model of the CIR developed can be used for the performance evaluation of a UWOC system.**

Keywords—UWOC, CIR Modeling, Monte-Carlo.

I. Introduction

Underwater Wireless Optical Communication (UWOC) is a promising technology useful in many areas such as defense, security, industry, search and rescue, and the environment. This is owing to the higher bandwidth, higher celerity, low latency, and low energy consumption, in contrast to the acoustic communication system[1]. However, the real-time experimentation and realization of UWOC systems in the field is cumbersome due to the harsh oceanic environment and rapid spatiotemporal variation of ocean optical properties. Therefore, theoretical evaluation of the performance of a UWOC system is highly essential before establishing a real-time communication system in the field. However, the estimation and modeling of power budget and channel impulse response (CIR) for various underwater environments are essentially the first step toward evaluating the performance of a UWOC system theoretically in terms of BER and channel availability. In the recent past, several works have been carried out in this direction to estimate the power budget and also Channel Impulse Response for the UWOC system theoretically[2], [3], [4], [5]. Further, some researchers have attempted to model the channel impulse response of the UWOC system[6], [7], [8]. However, most of these models are framed empirically based on the atmospheric CIR models. Therefore, some models are appropriate for clear oceanic waters, and some are appropriate for turbid waters. Moreover, no single comprehensive model is comprehensive including all the ocean-optical properties.

In the presented work, the authors have simulated the channel impulse response of various water oceanic types considering the realistic ocean-optical properties. The simulation was carried out using Monte-Carlo numerical simulation technique. The simulated CIRs for various water types are further modeled empirically to present a closed-form mathematical expression. The mathematical expression developed can be utilized to evaluate the performance of UWOC system in terms of bit error rate and link availability[9].

II. Estimation of Channel Impulse Response

The estimation of CIR requires solving the radiative transfer equation underwater given by (1).

$$\frac{dL(r,\theta,\phi,\lambda)}{dr} = -c(r,\lambda)L(r,\theta,\phi,\lambda) + \int_0^{2\pi}\int_0^{\pi} L(r,\theta',\phi',\lambda)\beta(r;\theta',\phi' \to \theta,\phi;\lambda)\sin\theta' d\theta' d\phi' + S(r,\theta,\phi,\lambda) \quad (1)$$

Where, $L(r,\theta,\phi,\lambda)$ is the radiance flux at a point, $c(r,\lambda)$ is the attenuation constant and $\beta(r;\theta',\phi' \to \theta,\phi;\lambda)$ is the scattering phase function. Further, θ, ϕ and λ are the polar angle, azimuthal angle, and wavelength, respectively.

The exact analytical solution of the radiative transfer equation underwater requires multiple approximations compromising the accuracy of the solution[10]. On the other hand, a numerical solution gives an approximate solution to the exact problem at hand. One such numerical method is the Monte-Carlo numerical simulation technique. The photon transport in water can be modeled using this technique with high accuracy and flexibility and without involving a series of assumptions and approximations[2], [4].

A. Monte-Carlo Numerical Simulation Technique

Despite its relatively low computation efficiency, the Monte-Carlo numerical technique was utilized for this work to estimate the channel impulse response along with the power budget. The flow diagram of the MC numerical technique is shown in Fig.1. For each simulation 10^6 photons were taken considering a trade-off between accuracy and computational burden. Since the MC simulation is a statistical technique, the simulation was repeated 10^3 times for each trans-receiver configuration and water type to ensure reliable and consistent results. The wavelength of light has been taken to be 532 nm because water has a minimum attenuation window in the blue-green spectrum. More detail on the Monte-Carlo simulation technique can be referred from Sahu and Shanmugam [2].

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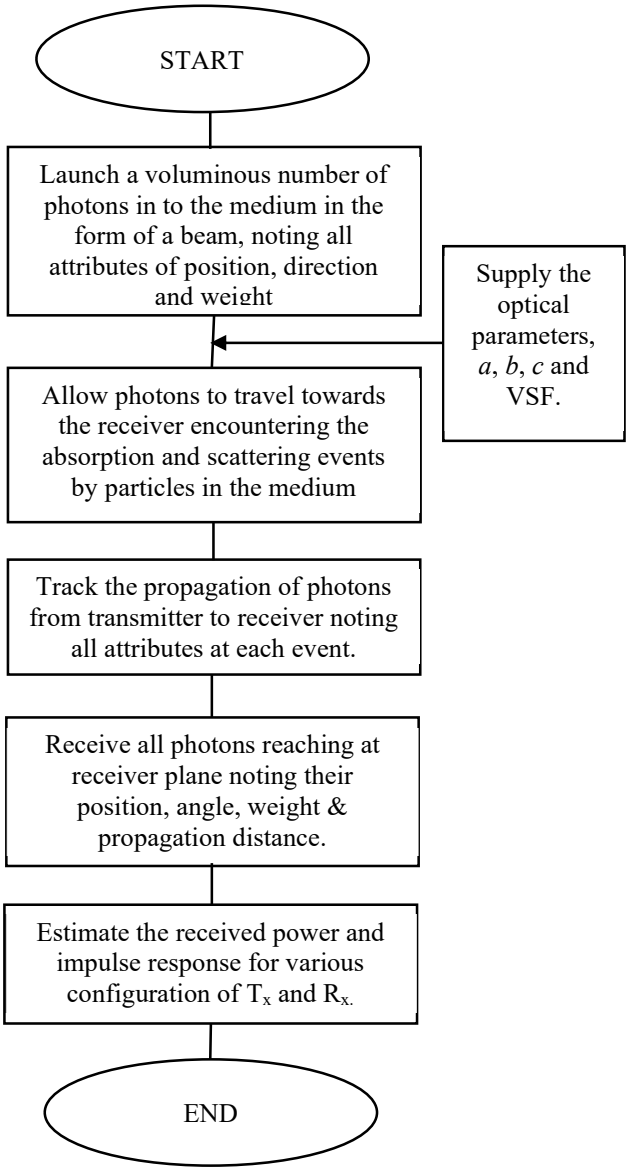


Fig.1. Flow diagram of the Monte-Carlo numerical simulation technique.

B. Simulation parameters

Various ocean-optical and system parameters were considered for estimating the CIR and Power budget as per Tables 1 and 2, respectively. The ocean-optical parameters were chosen carefully based on measured values provided in various literature[11]. The system parameters chosen were based on commercially available systems.

Table.1. Ocean Optical properties of various water types.

Type of Water	$a(m^{-1})$	$b(m^{-1})$	$c(m^{-1})$	ω_0
Clear Oceanic	0.113	0.037	0.15	0.25
Coastal Sea	0.181	0.219	0.4	0.55
Turbid Harbor	0.366	1.824	2.19	0.83

Where, a is absorption coefficient, b is scattering coefficient, c is the attenuation coefficient and ω_0 is the single scattering albedo.

Table.2. Source and receiver parameters for simulation.

Source		**Receiver**	
Beam radius (mm)	**Beam divergence angle (mrad)**	**Aperture (cm)**	**FOV (degree)**
2	1.5	10, 20, 50	45, 90, 180

Exhaustive simulation was carried out using the parameters tabulated in Table. 1 and 2 considering the SS phase function to estimate the scattering angle after each scattering event[12], [13]. The power received and impulse responses were estimated at various attenuation lengths (attenuation coefficient × physical distance) of 2 AL to 30 AL for costal and harbor water, and 2 AL to 20 AL for clear water. The simulated CIRs for various water types are further modeled empirically to obtain a closed-form mathematical expression for channel impulse response to fit all water types.

C. Impulse Response Modeling

The estimated CIR was modeled by leveraging double Gamma function similar to that for the CIR models of cloud[14].The closed-form expression for the double Gamma function is given in (2).

$$h(t) = C_1 \Delta t e^{-C_2 \Delta t} + C_3 \Delta t e^{-C_4 \Delta t}, \qquad (t > t_0) \qquad (2)$$

where, C_1, C_2, C_3, and C_4 are four parameters which can be found from the simulated data. And, $\Delta t = t - t_0$ where t is the time scale and $t_0 = {}^{L}/_{v}$ is the ballistic time, the minimum time required by the photons to arrive to the receiver.

The set of parameters (C_1, C_2, C_3, C_4) can be estimated from the Monte-Carlo simulation data by using the non-linear least square method as,

$$(C_1, C_2, C_3, C_4) = argmin\left(\iint [h(t) - h_{mc}(t)]^2 dt\right) \qquad (3)$$

where, $h(t)$ is the double Gamma functions model in (2) and $h_{mc}(t)$ is the simulated impulse response obtained using the Monte-Carlo method. The function $argmin(.)$ returns the argument of the minimum and (3) can be solved by curve fitting approach.

III. Results And Discussion

The simulated CIRs for clear water, coastal water and harbor water are shown in Fig. 1 for various attenuation lengths. For clear and coastal waters, the attenuation lengths were considered as AL = 4, 6 and 8, whereas, for harbor water the attenuation lengths 8, 10 and 12 are plotted. For all the CIRs plotted, the receiver aperture and FOV were 50 cm and 180° respectively, to visualize the maximum spreading of the impulse in the underwater channel and include the maximum multipath photons. The time in ns on the x-axis of the figure represents propagation time from transmitter to receiver normalized by the ballistic time of flight. A quantity delay spread (τ) can be defined as the period over which the CIR falls to -20 dB beneath its peak, to quantify the temporal spreading. The information is readily obtained from Fig. 1 that for lower attenuation lengths the CIR has a peak power at zero normalized time. This suggests that in case of clear and coastal waters and for lower values of attenuation lengths, most of the photons reach to the receiver ballistically and the multipath propagation of photons contributes very less to the

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pulse spreading. Such CIRs can be models by simple exponential functions instead of double Gamma function.

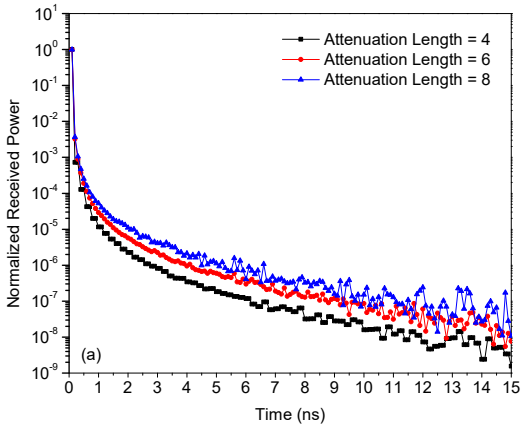


Fig. 2(a) CIR for clear water at AL = 4, 6 and 8.

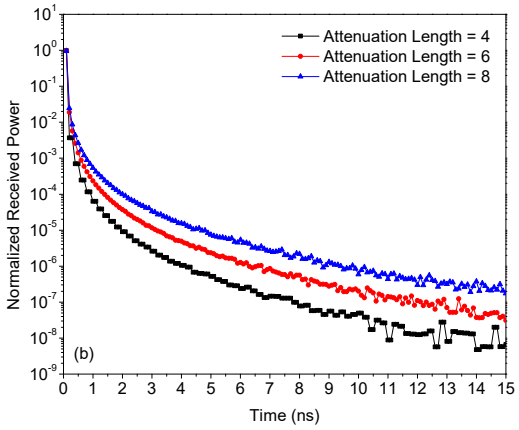


Fig. 2(b) CIR for coastal water at AL = 4, 6 and 8.

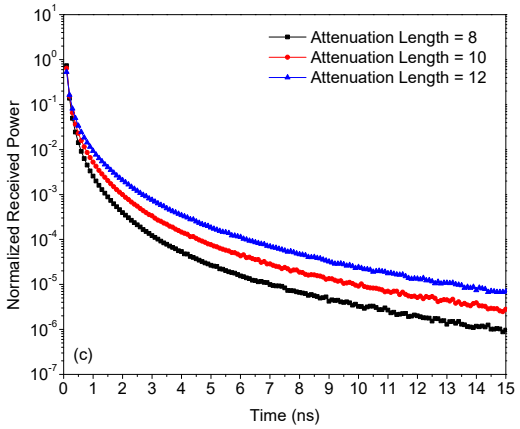


Fig. 2(c) CIR for harbor water at AL = 8, 10 and 12.

Although the CIRs for various water types do not possess the effect of multipath travel of photons at lower attenuation lengths, there is a significant effect of multipath propagation of photons on the CIRs at higher attenuation lengths. This can be observed in Fig.3 which shows the CIRs for coastal and harbor water at attenuation lengths of AL = 25, 27 and 30.

It can be observed from the figure that the normalized power peaks at normalized time values other than zero indicating the contribution of photons propagating in the multiple paths other than the straight ballistic path. Therefore, it is essential to model such peculiar behavior of CIRs especially for more turbid water and/or larger link lengths. The modeling was carried out using double Gamma function expressed by (2) and the coefficients were estimated using the least square method based on (3). The double Gamma function models for various attenuation lengths are plotted in the same figures to display the goodness of fit. In each case of fitting, it is ensured that the value of correlation coefficient (R^2) is approaches to one and the value of root mean square error (RMSE) approaches to zero.

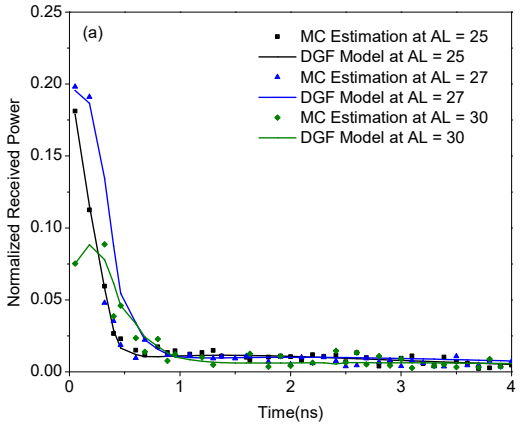


Fig. 3(a) CIR for coastal water at AL =25, 27 and 30.

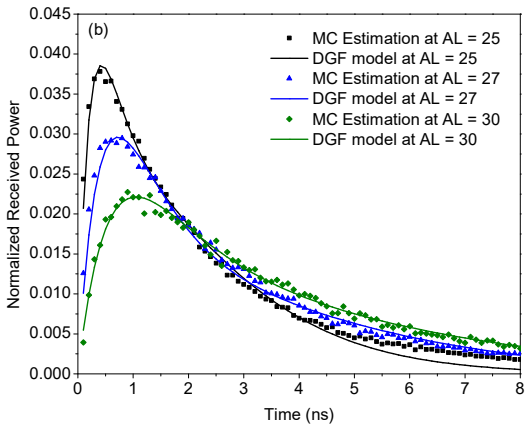


Fig. 3(b) CIR for harbor water at AL =25, 27 and 30.

The fitting coefficients along with the values of R^2 and RMSE are given in Table 3 and Table 4 for coastal and harbor water respectively, for various attenuation lengths. The values of R^2 and RMSE for each attenuation length show the goodness of fit. The fitting goodness decreases relatively at higher attenuation lengths due to the higher contribution of multiple scattered photons to the channel impulse response.

Table.3. Fitting coefficients, R^2 and RMSE for coastal water.

AL	C_1	C_2	C_3	C_4	R^2	RMSE
4	0.418	11.06	9280	68.68	0.9999	0.00098
6	0.409	10.27	2196	54.39	1	0.000002
8	0.3958	8.811	1589	51.35	1	0.000003
10	0.3642	7.262	1113	48.06	1	0.000003
12	0.3141	5.884	696.7	43.79	1	0.00032
14	0.2541	4.686	418.3	39.29	1	0.00015
16	0.1914	3.267	272.7	35.83	1	0.00014
18	0.1312	2.279	152.2	31.25	1	0.000084
20	0.0778	1.705	80.07	26.45	0.9984	0.00059
25	0.0223	0.7099	5.61	11.55	0.9102	0.00306
27	0.0136	0.4934	4.093	7.458	0.9815	0.00074
30	0.00656	0.3797	1.277	5.381	0.9837	0.00027

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Table.4. Fitting coefficients, R^2 and RMSE for harbor water.

AL	C_1	C_2	C_3	C_4	R^2	RMSE
4	1.143	7.196	109.3	27.79	0.9999	0.000406
6	0.9904	5.906	73.31	25.25	0.9998	0.00068
8	0.8373	4.831	45	22.44	0.9999	0.00101
10	0.6396	3.81	24.12	19.15	0.9983	0.00138
12	0.4466	2.953	12.23	15.91	0.9961	0.00163
14	0.2889	2.244	5.439	12.33	0.9923	0.001751
16	0.1632	1.615	2.162	8.783	0.989	0.00138
18	0.09015	1.17	0.887	5.95	0.9568	0.00138
20	0.03437	0.7167	0.231	2.907	0.9929	0.00077
25	0.01829	0.5326	0.114	1.881	0.9959	0.000506
27	0.00931	0.384	0.052	1.196	0.9955	0.000429
30	1.143	7.196	109.3	27.79	0.9999	0.000406

The variation of the estimated values of fitting coefficients (C_1, C_2, C_3, C_4) with attenuation length were further analyzed. Figure. 4 and 5 show the variation of all fitting coefficients with attenuation lengths for coastal water and harbor water, respectively.

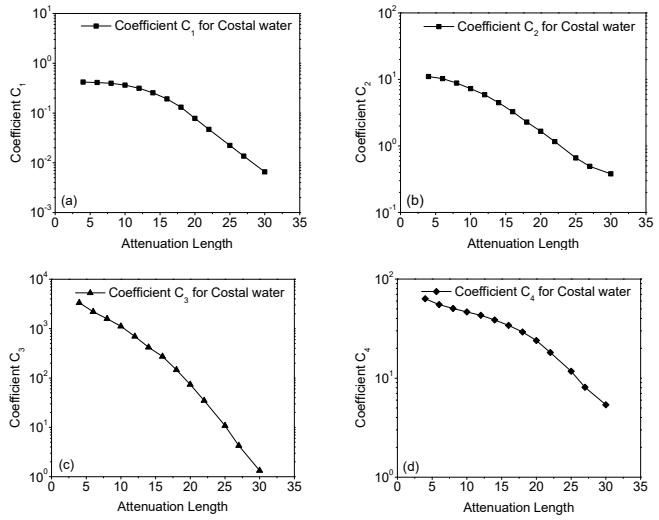


Fig.4. Variation of fitting coefficients (a) C_1 (b) C_2 (c) C_3 and (d) C_4, with attenuation lengths for coastal water.

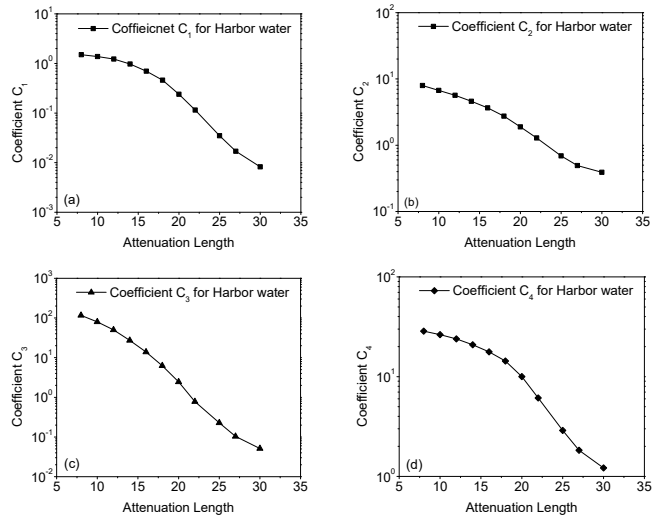


Fig.5. Variation of fitting coefficients (a) C_1 (b) C_2 (c) C_3 and (d) C_4, with attenuation lengths for harbor water.

It can be observed from the figures that the variation of each coefficient with attenuation lengths follows a similar pattern. Therefore, the coefficient can further be modeled as a function of attenuation lengths to obtain a closed-form mathematical expression for CIR including attenuation length.

IV. Conclusion And Future Work

In the presented work comprehensive simulation was carried out to obtain the channel impulse response of an UWOC system for various water types using Monte-Carlo numerical simulation technique. The information is directly observable that the CIR gets the peak power at a time separated from the zero normalized time for higher turbidity and/or larger link lengths. This behavior was attributed to the multiple paths followed by the photons while traveling from transmitter to receiver owing to multiple scattering in a more turbid environment. The CIR obtained for various water types were further modeled by using double Gamma function to obtain a closed form mathematical expression. The coefficients for the fitting were obtained using the least square method and the goodness of fit was ensured by confirming the values of R^2 and RMSE to approach to their ideal values of one and zero, respectively. The obtained closed form mathematical expression for impulse response shall be useful to assess the performance of a UWOC system in terms of BER and link availability.

In future, the authors plan to include the other variables such as attenuation lengths, beam divergence angle, receiver aperture and FOV in the CIR model to get a more inclusive closed-form mathematical expression.

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